MACHINE LEARNING

Unit 3: Supervised Learning

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Introduction to Machine Learning

Machine Learning (ML) enables systems to learn patterns from data — improving over time without being explicitly programmed for every task.

Spam Detection

Emails classified as spam or legitimate using learned patterns

Medical Diagnosis

Predicting diseases from patient records and test results

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Personalized product and content suggestions for users

Types of Machine Learning

Supervised Learning

Trains on **labeled data** to map inputs to known outputs.
e.g., spam detection, image classification

Unsupervised Learning

Finds **hidden patterns** in unlabeled data.
e.g., customer clustering, anomaly detection

Semi-Supervised

Uses a **mix of labeled and unlabeled** data to reduce labeling cost.
e.g., web content classification

Reinforcement Learning

Learns via **rewards and penalties** through trial and error.
e.g., game playing, robotics

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What is Classification?

Classification is a supervised learning task that assigns input data to predefined categories based on learned patterns.

- Binary: Spam vs. Not Spam
- Multi-class: Cat, Dog, Bird
- Medical: Benign vs. Malignant

Classification Algorithms

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Tree-based model that splits data using attribute tests to reach class predictions

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K-Nearest Neighbors

Classifies by majority vote among the K closest labeled training points

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SVM

Finds the optimal hyperplane that maximizes the margin between classes

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Random Forest

Ensemble of decision trees that votes for improved accuracy and robustness

Decision Tree

A **Decision Tree** is a flowchart-like model that recursively splits data based on attribute values to reach a class prediction.

Root Node

Top node; best splitting attribute

Internal Node

Tests an attribute, branches further

Leaf Node

Final class label output

Splitting Measures

Entropy:

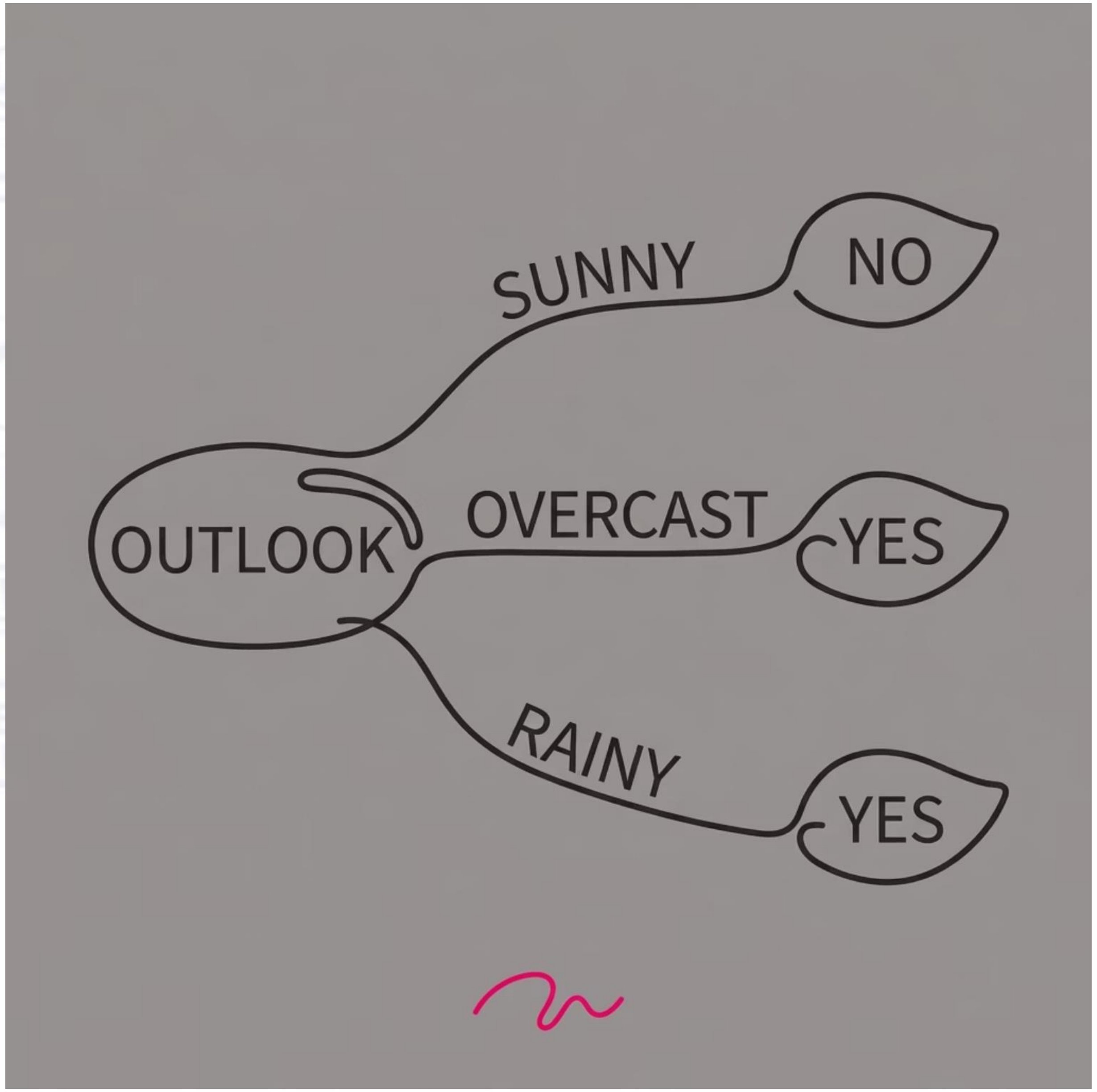
Measures impurity in a dataset. **Entropy = 0** means the set is perfectly pure (one class only).

$$H(S) = -p_1 \log_2 p_1 - p_2 \log_2 p_2$$

Information Gain

Reduction in entropy after splitting on an attribute. The **highest gain** indicates the best split.

$$IG(S, A) = H(S) - \sum_{v} \frac{|S_v|}{|S|} H(S_v)$$



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Problem 1 — Build the Tree

Outlook	Play
Sunny	No
Sunny	No
Overcast	Yes
Rainy	Yes
Rainy	Yes

Root = Outlook → Sunny: No · Overcast: Yes · Rainy: Yes

Problem 2 — Best Attribute

Outlook	Temp	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rainy	Cool	Yes

Best split = Outlook (highest Information Gain)

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Input: Outlook = Sunny → **Prediction = No**

K-Nearest Neighbors (KNN)

KNN is a lazy, instance-based learner. It stores all training data and classifies new points by **majority vote** among the K closest neighbors.

1. Choose K

   Select the number of neighbors

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   Euclidean distance to all training points

3. Find Neighbors

   Identify K closest points

4. Majority Vote

   Assign the most frequent class

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Training Data

Point	Class
(2, 2)	A
(4, 4)	A
(1, 1)	B
(5, 5)	B

Distances from (3, 3)

Neighbor	Distance
(2, 2) → A	1.41
(4, 4) → A	1.41
(1, 1) → B	2.83

3 Nearest: **A, A, B** → Majority = **Class A**

THANK YOU